Structure of the model (v7)

Seventeen actors, simulated year by year, 2026 to 2075, as a Monte Carlo over all uncertain parameters. Labels in brackets refer to the assumptions in Section 4 of the paper.

Scenario tree around the actor simulation

Earlier stages (US and China, regimes still staffed by humans): automation race → collapse of broad economic participation → broad repression.
Competing end points: AI takeover not directed by rulers, non-AI global catastrophe (a run that hits one first is not counted).
Counterfactual: if the US becomes broadly repressive first, all seventeen actors continue from a world where the US is autocratic at closure.

Inside each actor: US, China, Japan, South Korea, Taiwan, Western EU, Eastern EU, UK, Canada-Australia-NZ, Russia and Central Asia, Gulf, rest of MENA, India, rest of South Asia, Southeast Asia, Latin America, sub-Saharan Africa. Each actor is modeled as one state with one ruling coalition.

1 AI and robots

AI capability follows the measured task-horizon trend and can speed up (A1). An actor's own chips and energy add capability; others lag and catch up. Robots build robots, limited by energy, chips, mining and world inputs (A2, B6).

2 Automation and closure

Nine economic segments automate. A dedicated automated core loop can be built beside the old economy.
Human-dependence index: share of 2026 labor still needed.
CLOSURE when it falls below 20%.

3 Force

Security automates first (A3). Once fielded at scale, refusal by soldiers and police stops mattering (A4).
Military power = spending x output x automated share. Power inside the ruling group = control of that force (A6).

4 Democracy

V-Dem liberal democracy score. Falls as labor and revolt leverage fall. Below 0.30 = breakdown. Sets the decision rule in box 7.

5 The public's leverage

Labor: people can stop working.
Revolt: people can fight.
Enforcers: soldiers can refuse.
All three fall with closure (box 2) and automated force (box 3).

6 Grievance and unrest

Displacement and provision cuts (no safety net by default, A5) → grievance → hostility → yearly insurgency hazard → repression. Provision only partly buys peace. Allied autocracies may send forces to put down an uprising (box 10).

threat

7 The ruling coalition decides

Every year once the actor reaches closure. Up to 21 members (autocracy) or a 51-person elite (democracy), drawn from the state, the military and the owners of AI and industry. Each has, correlated through seven drives:
- moral cost of ordering killing (lower at the top in autocracies, B1)
- concern for the public, discount rate
- share of automated force (box 3)

Each member scores six options:

\+ public's welfare x concern
− cost of provision (falls with growth)
− threat of revolt and sabotage
− latent threat: what survivors could later become, even when quiet
− outside pressure if enforceable (B3)
\+ preference for an emptier planet, only under a depopulation ideology
− own moral cost of killing or neglect

Decision rule depends on regime:

democracy: 3 to 5 veto players
oligarchy: force-weighted supermajority
personalist: leader decides, inner circle may veto

Inside the group:

purges and counter-coups (B2). A depopulation ideology can spread, faster in war and unrest. Brakes: fear of being next, leader mortality, protective doctrine.

fear of survivors (A11, B4)

8 Options (mild → harsh)

1 serve the public
2 live off rents
3 keep the status quo
4 warehouse (minimal provision)
5 neglect
6 depopulate

9 How deaths happen

Ordered depopulation: automated force, economic strangulation, abstract means of mass killing (A8)
Chosen, with no order to kill: lethal neglect, partly through a deniable passive channel
War, up to nuclear exchange (A9); attacks on weaker actors (B5)
Total loss only: despair, crackdowns

What stops a campaign

a successful revolt, a repelled attack, or nuclear retaliation

10 Between actors

Alliances: defense commitments that deter attacks and extend nuclear deterrence (B5), draw allies into wars, enforce outside pressure (B3) and ease trade in robot inputs (B6). Allies never attack each other. Atrocities break ties; autocracies form new ones.

Allied autocracies send forces to help put down uprisings. Foreign troops do not defect, so revolts succeed less often. Democracies do not send forces. The helper's cost to its own defenses is not modeled.

The US as of October 2026: weak commitments to Europe, Canada and Oceania, middling to Japan and Korea, partial alignment with Russia (no defense). Ties do not recover after a US return to democracy. Small US-Europe war hazards.